LF-mapping and pattern matching using BWT Proofs

Arun Konagurthu

Notations

$S[1 \ldots n]$	String of n characters including the terminal \$.
M	Sorted $n \times n$ cyclic permutation matrix of string S.
$M_{\text{row}(i)}$	The ith row of the matrix M.
$M_{\text{col}(i)}$	The ith column of the matrix M.
$L[1 \ldots n]$	Last column of M, or, equivalently, $M_{\text{col}(n)}$. The Burrows-Wheeler Transform (BWT) of the string S.
$F[1 \ldots n]$	First column of M, or, equivalently, $M_{\text{col}(1)}$. By the definition of M, $F[1 \ldots n] = \texttt{sort}(L[1 \ldots n])$.
$\text{Rank}(x)$	The rank (i.e., the position) of the first observation of any character x in F.
$\text{nOcc}(x, L[\text{'range'}])$	The number of occurrences of a character x in any prescribed 'range' of L. Below, the 'range' can sometimes be a righ-open interval $L[\cdot, \cdot)$ and sometimes closed $L[\cdot, \cdot]$.
$\texttt{rcylshift}(M_{\text{row}(i)}, 1)$	Right cyclic shift of $M_{\text{row}(i)}$ by 1 position.
$\alpha \prec \beta$	Lexicographic precedence between two string α and β. That is, the string α is lexicographically smaller than the string β.

Theorem 1 (LF-Mapping)**.**
For all $1 \leq i \leq n$, $L[i] \xrightarrow{\textit{maps to some}} F[p]$ *in* M, *where* $p = \textit{Rank}(L[i]) + \textit{nOcc}(L[i], L[1 \ldots i])$.

Proof. Let $L[i] = x$. As a consequence of $F[1 \ldots n] = \texttt{sort}(L[1 \ldots n])$, it follows that

$$\text{Rank}(x) \leq p \leq \text{Rank}(x) + \text{nOcc}(x, L[1 \ldots n]) - 1.$$

Further, from Lemma 2 and Corollary 3 (see below), it follows that the relative-order in which all the $k \geq 1$ instances of the character x (one instance of which is observed at $L[i]$) will remain preserved in F, and these instances of x have to appear *en bloc* starting at the position $\text{Rank}(x)$.

$\text{nOcc}(x, L[1 \ldots i))$ gives the number of occurrences of x before position i in L. This implies that, in F, the specific instance of $x = L[i]$ will have to follow after $\text{nOcc}(x, L[1 \ldots i))$ instances that start at $\text{Rank}(x)$.

$$\boxed{\implies p = \text{Rank}(x = L[i]) + \text{nOcc}(L[i], L[i \ldots i)).}$$

$\square$

Lemma 2. *For all $1 \leq i_1 < i_2 \leq n$ such that $L[i_1] = L[i_2] = x$,*

$$M_{row(i_1)} \prec M_{row(i_2)} \implies \texttt{rcylshift}(M_{row(i_1)}, 1) \prec \texttt{rcylshift}(M_{row(i_2)}, 1)$$

Proof. Assume that $M_{\text{row}(i_1)}$ and $M_{\text{row}(i_2)}$ define two distinct cyclic permutations that start at j_1 and j_2 (respectively) in S. This implies:

$$\begin{aligned} M_{\text{row}(i_1)} &= \underbrace{S[j_1]S[j_1+1]S[j_1+2]\ldots S[n]S[1]S[2]\ldots S[j_1-2]}_{\text{denote this prefix as } \alpha}\underbrace{S[j_1-1]}_{L[i_1]=x}. \\ M_{\text{row}(i_2)} &= \underbrace{S[j_2]S[j_2+1]S[j_2+2]\ldots S[n]S[1]S[2]\ldots S[j_2-2]}_{\text{denote this prefix as } \beta}\underbrace{S[j_2-1]}_{L[i_2]=x}. \end{aligned}$$

Since $i_1 < i_2$, by the definition of M, we have $M_{\text{row}(i_1)} \prec M_{\text{row}(i_2)}$. Since the last characters of these two rows are the same ($L[i_1] = L[i_2] = x$), for the precedence of $M_{\text{row}(i_1)}$ over $M_{\text{row}(i_2)}$ in M to hold, it has to be that $\alpha \prec \beta$.

When the rows $M_{\text{row}(i_1)}$ and $M_{\text{row}(i_2)}$ are shifted right cyclically by 1 position, their last characters will circle to the first positions (respectively) in their corresponding rows in M:

$$\begin{aligned} \texttt{rcylshift}(M_{\text{row}(i_1)}, 1) &= \underbrace{S[j_1-1]}_{L[i_1]=x}\underbrace{S[j_1]S[j_1+1]S[j_1+2]\ldots S[n]S[1]S[2]\ldots S[j_1-2]}_{\text{denote this prefix as } \alpha}. \\ \texttt{rcylshift}(M_{\text{row}(i_2)}, 1) &= \underbrace{S[j_2-1]}_{L[i_2]=x}\underbrace{S[j_2]S[j_2+1]S[j_2+2]\ldots S[n]S[1]S[2]\ldots S[j_2-2]}_{\text{denote this prefix as } \beta}. \end{aligned}$$

In this cyclically shifted rows, the first characters. Thus,

$$\boxed{\alpha \prec \beta \implies \texttt{rcylshift}(M_{\text{row}(i_1)}, 1) \prec \texttt{rcylshift}(M_{\text{row}(i_2)}, 1).}$$

$\square$

Corollary 3. *If there are k instances of some character x at positions $\{i_1, i_2, \ldots i_k\}$ in L, then*

$$\texttt{rcylshift}(M_{row(i_1)}, 1) \prec \texttt{rcylshift}(M_{row(i_2)}, 1) \prec \ldots \prec \texttt{rcylshift}(M_{row(i_k)}, 1).$$

Theorem 4 (Pattern-matching using BWT – Update rules)**.**
Let some string α of length $|\alpha|$ appear ≥ 1 times in S. Let $[sp, ep]$ define the range of rows in M such that the first $|\alpha|$ columns of M have the same characters as α. Then, all occurrences of the string $x\alpha$ (i.e., α left-extended by the character x), if it appears as a substring of S one or more times, will have to appear as the first $|\alpha| + 1$ columns of M within the range of rows identified by $[sp', ep']$, where

$$\begin{aligned} sp' &= Rank(x) + nOcc(x, L[1 \ldots sp)), \\ ep' &= Rank(x) + nOcc(x, L[1 \ldots ep]) - 1. \end{aligned}$$

Proof. The range $[\mathrm{sp}, \mathrm{ep}]$ defines the block of rows in M such that the first $|\alpha|$ columns of M have the same characters as α.

By the property that the column L precedes column F in M, the string $x\alpha$ (i.e., the left-extension of α by the character x) appears in S if and only if $x \in L[\mathrm{sp} \ldots \mathrm{ep}]$.

If x appears in the the range $L[\mathrm{sp} \ldots \mathrm{ep}]$, define $L[i] = x$ as x's first occurrence in that specific range and $L[j] = x$ as x's last occurrence in that specific range, where $\mathrm{sp} \leq i \leq j \leq \mathrm{ep}$.

Using Theorem 1, we know that $L[i] \xrightarrow{\text{maps to some}} F[\mathrm{sp}']$ in M, where

$$\mathrm{sp}' = \mathrm{Rank}(x) + \mathrm{nOcc}(x, L[1 \ldots i)).$$

By the very definition of $L[i] = x$ being the 'first' occurrence of x in the range $L[\mathrm{sp} \ldots \mathrm{ep}]$, it has to be true that

$$\mathrm{nOcc}(x, L[1 \ldots i)) = \mathrm{nOcc}(x, L[1 \ldots \mathrm{sp})).$$

This holds because the range in which the occurrences are being counted is exclusive of the occurrence at position i.

This proves the update rule required to the start point upon left-extension of α by character x is:

$$\boxed{\mathrm{sp}' = \mathrm{Rank}(x) + \mathrm{nOcc}(x, L[1 \ldots \mathrm{sp}))}$$

Again, using Theorem 1, we know that $L[j] \xrightarrow{\text{maps to some}} F[\mathrm{ep}']$ in M, where

$$\mathrm{ep}' = \mathrm{Rank}(x) + \mathrm{nOcc}(x, L[1 \ldots j)).$$

By the definition of $L[j] = x$ being the last occurrence of x in the range $L[\mathrm{sp} \ldots \mathrm{ep}]$, it has to be true that

$$\mathrm{nOcc}(x, L[1 \ldots j)) = \mathrm{nOcc}(x, L[1 \ldots \mathrm{ep}]) - 1.$$

This holds true because the right hand side of the above relationship counts all occurrences of x in the range $L[1 \ldots \mathrm{ep}]$ (n.b. inclusive). Since $j \leq \mathrm{ep}$, r.h.s would include the specific instance of $L[j] = x$, which the left hand side needs to omit. Thus, the r.h.s has to compensate this by subtracting 1.

Therefore, this proves the update rule required to the end point upon left-extension of α by x:

$$\boxed{\mathrm{ep}' = \mathrm{Rank}(x) + \mathrm{nOcc}(x, L[1 \ldots \mathrm{ep}]) - 1}$$

From Lemma 2 and Corollary 3, it follows that the range of rows $[\mathrm{sp}', \mathrm{ep}']$ is exhaustive, in the sense that the no other rows of M beyond those defined by $[\mathrm{sp}', \mathrm{ep}']$ will contain $x\alpha$ in the first $|\alpha| + 1$ columns of M. $\square$

Corollary 5. *When $x\alpha$ does not appear as a substring in S, then Theorem 4 results in the evaluation of $ep' = sp' - 1$, that defines a null range. In other words, number of occurrences will be $ep' - sp' + 1 = 0$.*

Corollary 6. *Using Theorem 4 and Corollary 5, any pattern $pat[1 \ldots m]$ can be searched using $L[1 \ldots n]$ (i.e., BWT of S) as a search index, starting with $\alpha = \varnothing$ (empty/null string) and $[sp, ep] = [1, n]$ and iteratively extending the search backwards one character at a time.*